

Wellington Santos

Staff Data Engineer

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PROFESSIONAL SUMMARY

Staff Data Engineer with **8+** years across e-commerce, sports commerce, and healthcare, building production data infrastructure and analytics platforms. Expert in **Snowflake**, medallion architecture, and modern data transformation using **dbt** and **Coalesce**. Proven track record designing scalable pipelines, optimizing warehouse performance, implementing data governance, and enabling AI-ready data products for enterprise SaaS environments.

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, Bash, HTML, CSS

Data & ML: Snowflake, dbt, Coalesce, Apache Airflow, Apache Spark, Pandas, NumPy, Fivetran, Kafka, Tableau, Looker, Metabase

Backend: Node.js, Express, NestJS, FastAPI, Django, Spring Boot, PostgreSQL, MongoDB, RabbitMQ

Database: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB, Cassandra

Testing: Jest, Pytest, Cypress, Selenium, Playwright, k6

Cloud & DevOps: AWS, AWS Lambda, S3, EC2, RDS, SQS, SNS, CloudWatch, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins

Others: RESTful API, GraphQL, gRPC, WebSocket, Git, GitHub, Jira, Confluence, Data Governance, RBAC, Data Lineage, PII Classification, Data Contracts, Vector Search, RAG Pipelines, LangChain, Observability, Prometheus, Grafana

EXPERIENCE

Fanatics
Staff Data Engineer

Oct 2023–Present

- Led the design and deployment of a medallion architecture spanning Bronze, Silver, and Gold layers across the sports commerce platform, implementing **Snowflake** warehouse tuning strategies that reduced query latency from **1,200ms to 380ms** while cutting compute costs by **28%** through strategic clustering and partition elimination.
- Spearheaded the migration of legacy batch ETL pipelines to **Fivetran** and **Coalesce** transformation logic, establishing semantic layer data products representing betting markets, collectibles inventory, and fan engagement metrics that powered **15+ Tableau** dashboards serving GTM and finance teams.
- Built end-to-end data observability practice by implementing freshness monitoring, anomaly detection thresholds, and SLA tracking in **Prometheus** and **Grafana**, reducing mean-time-to-recovery for data incidents from **4 hours to 45 minutes** and catching pipeline failures before they affected downstream analytics.

- Designed and documented data contracts across product, payments, and subscription systems, enforcing PII classification, RBAC policies, and lineage tracking in the governance layer — enabling compliance audits to complete in **1 day** instead of **3 weeks** of manual reconciliation.
- Mentored two senior engineers through hands-on code review and design pairing on complex query optimization and Coalesce module architecture, establishing SQL best practices and medallion layer standards that the team now consistently applies.
- Evaluated and led the technical rationale for Snowflake Cortex as the semantic layer foundation for AI-ready data products, including vector search and embedding retrieval patterns to support LLM and RAG agent use cases across business intelligence.
- Owned the fan engagement data product pipeline, integrating ticketing, collectibles marketplace, and sportsbook event streams into a unified **Snowflake** semantic model that reduced ad hoc analytics requests by **45%** and enabled self-service exploration for product stakeholders.
- Partnered with Data Science and Product teams to curate enriched, contextually grounded datasets for embeddings-based search and recommendation engine training, establishing governance checkpoints and documentation standards for AI-consumed data.

Walmart

Sep 2022–Oct 2023

Senior Data Engineer

- Drove the implementation of a scalable data ingestion and transformation platform serving e-commerce, advertising, and fintech domains using **Snowflake**, **Apache Spark**, and **Airflow**, processing **500M+ daily events** across checkout, fulfillment, and marketplace subsystems.
- Architected dimensional data models and fact tables for retail analytics using **dbt**, designing the transformation layer to support inventory forecasting, demand sensing, and revenue attribution across Walmart.com, Sam's Club, and Walmart Connect advertising platform.
- Led performance tuning initiatives on high-concurrency **Snowflake** warehouses, optimizing query patterns for Tableau dashboards serving finance and merchandising teams — eliminated resource contention by implementing result caching, micro-partition pruning, and materialized views that improved dashboard load times by **52%**.
- Implemented data governance policies including PII masking rules, role-based access control, and data lineage documentation for a sensitive payroll and benefits dataset, ensuring GDPR and SOC 2 compliance across **18 interconnected pipelines**.
- Collaborated with the BI team to design and ship semantic tables in **Snowflake** representing key business entities — orders, customers, products, payments — that reduced Tableau custom SQL requests by **60%** and accelerated time-to-insight for merchandising analysts.
- Supported three junior data engineers through code review and architectural guidance on pipeline design, query optimization, and medallion layer implementation — establishing team standards that improved deployment reliability to **99.7%** uptime.
- Designed and maintained data observability monitors for **40+ critical pipelines**, capturing freshness SLAs, record count anomalies, and null value thresholds in **Prometheus** — enabling the team to proactively surface data quality issues before impact.

E-POT

Jun 2018–July 2022

Full Stack Data Engineer

- Built and maintained an end-to-end data platform for healthcare IT operations, ingesting electronic health record systems, laboratory order management, and financial transaction logs into a **PostgreSQL**-backed warehouse, then materializing analytics tables in **Snowflake** for compliance reporting.
- Designed medallion-pattern data transformations using **Python** and **SQL** across Bronze (raw ingest), Silver (cleaned, deduplicated), and Gold (business-ready) layers for medical billing, pharmacy operations, and patient demographic datasets — reducing manual data reconciliation effort by **35%**.
- Implemented **Apache Airflow** orchestration for **25+ daily jobs** spanning healthcare data pipelines, payment processing workflows, and inventory synchronization, implementing retry logic and SLA monitoring that kept average pipeline execution within SLA **98.2%** of the time.
- Created semantic models and **dbt** transformations for key healthcare business entities — patient encounters, clinical orders, pharmacy fills, billing claims — that fed Looker dashboards used by operations and finance teams to track KPIs and compliance metrics.
- Optimized query performance on legacy **PostgreSQL** systems by adding strategic indexing, query plan analysis, and materialized view design, cutting dashboard refresh times from **8 minutes to 90 seconds** for the finance reporting system serving **50+ daily users**.
- Partnered with security and compliance teams to implement data governance controls including PII detection and masking rules for protected health information, lineage documentation, and role-based access policies aligned with HIPAA requirements.
- Maintained data quality assurance by building **Pytest**-based validation frameworks that detected anomalies in patient record counts, billing amounts, and pharmacy order volumes — catching data integrity issues early and preventing downstream analytics errors.

EDUCATION

University of Tokyo

Apr 2014–Mar 2018

Bachelor of Science in Computer Science